Migraine prophylaxis 2024 evidence?

Evidence searched on 2026-10-14 (PubMed, ClinicalTrials.gov, FDA drug labels, web)

Clinical summary

Bottom line The 2024 evidence supports **oral prophylactic drugs (β-blockers, amitriptyline, topiramate, flunarizine) and monoclonal antibodies targeting CGRP** for episodic migraine prophylaxis, with reductions in monthly migraine days (e.g., 1.2–1.4 days vs. placebo). **Ultrasound-guided greater occipital nerve block (US-GONB)** also shows short-term efficacy in reducing pain intensity and headache frequency, though its role remains adjunctive. **Phenotype-guided approaches (e.g., venlafaxine for vestibular migraine)** are under investigation but not yet standardized.

Most important recent finding Ultrasound-guided greater occipital nerve block (US-GONB) significantly reduced monthly headache frequency by **9.12 days** (MD = −9.12, 95% CI [−11.65, −6.58]) and pain intensity (MD = −3.48 points) in a systematic review/meta-analysis of 12 studies (658 patients). [1]

Immediate context Established oral prophylactic agents (β-blockers, tricyclics, anticonvulsants) and CGRP monoclonal antibodies remain the cornerstone for episodic migraine prophylaxis, with clear efficacy in reducing migraine days. **US-GONB is emerging as a non-pharmacological option** but lacks long-term data or FDA approval

Top treatment drugs (established first, then the most notable latest drug)

Drug	Indication named in the FDA label	Source
Valproic Acid	"For prophylaxis of migraine headaches, valproate is contraindicated in women who are pregnant and in women of childbearing potential who are not using effective contraception."	DailyMed

Latest findings

Ultrasound-guided greater occipital nerve block (US-GONB) (2026, advance publication, *experimental evidence*) A systematic review and meta-analysis of 12 studies (658 patients) found that US-GONB significantly reduced pain intensity, monthly headache frequency, headache duration, and analgesic use. The proximal injection approach showed numerically larger reductions in headache frequency than the distal approach, though adverse events (e.g., loss of pinprick sensation, dizziness) were mild and transient [1].

Melatonin (2026, *emerging evidence*) While less potent than amitriptyline, melatonin was well-tolerated and may serve as a supplementary option during preventive agent titration [5].

Rimegepant for short-term prevention (2026, *emerging evidence*) An observational study of 425 U.S. adults found that women were more likely to use rimegepant for short-term prevention due to doctor recommendation, frequency, and severity of attacks. Both sexes reported high satisfaction (>=73%) with the treatment, though men showed numerically higher satisfaction with one-time use [6].

Dietary supplements (magnesium, CoQ10, riboflavin, etc.) (2025, *emerging evidence*) A systematic review and meta-analysis of 22 trials showed that magnesium, CoQ10, riboflavin, and vitamin D reduced migraine frequency, severity, and monthly migraine days compared to placebo. Probiotics also decreased attack frequency and severity, though effects varied by supplement [8].

Rimegepant for preventive migraine treatment (2025, *established evidence*) Adverse events were mild, and no liver toxicity signals were observed [9].

Current evidence

Approach	Evidence level	What the evidence shows	Reference
CGRP monoclonal antibodies (e.g., galcanezumab, erenumab, fremanezumab, eptinezumab)	Evidence-supported but limited	High-confidence evidence supports reductions in headache frequency for episodic and chronic migraine; network meta-analysis ranked eptinezumab 300 mg and fremanezumab monthly as top choices for headache/migraine days	[10]
Oral CGRP receptor antagonists (e.g., atogepant)	Established	Moderate-confidence evidence supports efficacy in reducing migraine days per month (0.7–2.4 fewer days) for episodic and chronic migraine	[11]
OnabotulinumtoxinA (Botox)	Evidence-supported but limited	High-confidence evidence supports reductions in headache frequency for chronic migraine	[10]
Topiramate	Established	High-confidence evidence shows reductions in migraine days (1.2–1.4 fewer days/month vs. placebo) for episodic migraine; moderate-confidence for chronic migraine	[11]
Melatonin	Emerging	Meta-analysis of RCTs (n=788) shows reduced migraine attack duration (MD -4.98 h) and headache days (MD -1.54 days) vs. placebo, with a favorable safety profile	[5]
Magnesium, CoQ10, riboflavin, vitamin D, probiotics, alpha-lipoic acid	Emerging	Meta-analyses report reductions in migraine frequency, severity, and monthly migraine days (e.g., magnesium: MD -2.51 attacks; CoQ10: MD -1.73 frequency) vs. placebo	[8]
Ultrasound-guided greater occipital nerve block (US-GONB)	Experimental or preliminary	Meta-analysis (n=658) shows significant reductions in pain intensity (MD -3.48 points), monthly headache frequency (MD -9.12 days), and analgesic use vs. placebo; proximal injections may offer larger reductions	[1]
Rimegepant (preventive use)	Evidence-supported but limited	Not retrieved	[9]

Treatment Drug Landscape

Drugs used for treatment of this condition

Approved: an FDA label names this condition. Guideline-recommended: a guideline organisation's page recommends the drug. Used in practice: a retrieved review or page describes the drug for this condition, with no FDA label or guideline for it (see the category in Key evidence). Doses appear only where an FDA label or source gives them for the use shown; individual dosing is a prescriber's decision.

Drug	Indication	Dose / route (if sourced)	Key evidence
Valproic Acid	"For prophylaxis of migraine headaches, valproate is contraindicated in women who are pregnant and in women of childbearing potential who are not using effective contraception."	Route: Oral. No dosing subsection for this condition; the label's general dosing (not specific to this use): "2 DOSAGE AND ADMINISTRATION Valproic acid oral solution is intended for oral administration. (2.1) Simple and Complex Absence Seizures: Start at 10 to 15 mg/kg/day, increasing at 1 week intervals by 5 to 10 mg/kg/week until seizure control or limiting side effects (2.1) Safety of doses above 60 mg/kg/day is not established (2.1 , 2.2) 2.1 Epilepsy Valproic acid oral solution is intended for oral administration. Patients should be informed to take valproic acid oral solution every day as prescribed. If a dose is missed it should be taken as soon as possible, unless it is almost time for the next dose. If a dose is skipped, the patient should not double the next dose. (the label continues)"	FDA label. DailyMed label FDA-approved for the use shown; label effective 2024-05-27.

Other drugs: used for symptoms (not established treatments of this condition)

Drugs that a retrieved review, guideline or web page describes as used for the condition, its symptoms or co-occurring conditions, for which no FDA label for this use was retrieved. Drugs still being tested in trials are listed in the Clinical trials section below.

Drug	Indication	Dose / route (if sourced)	Key evidence
Topiramate	50% reduction in monthly migraine days	Not stated in the retrieved source for this use	"Compared to placebo, Botulinum toxin A demonstrated the most significant effect in reducing the monthly migraine days for CM patients (MD = 3.88, 95% CI 0.48, 7.28); in terms of improving the response rate by a 50% reduction in monthly migraine days, Topiramate (RR = 50.06, 95% CI 3.18, 787.30) was the most effective; there was no statistically significant difference between all preventive drugs and placebo in ..." [2]
Eptinezumab	preventive treatment for CM	"Dec 7, 2025 · Eptinezumab Dosing and Treatment Approach for Migraine Prevention For patients with migraine who have failed other preventive treatments, initiate eptinezumab 100 mg IV every 3 months, with assessment of efficacy only after 3-6 months of treatment, and consider dose escalation to 300 ..." source (from a retrieved page, not an FDA label)	"Compared to placebo, Botulinum toxin A demonstrated the most significant effect in reducing the monthly migraine days for CM patients (MD = 3.88, 95% CI 0.48, 7.28); in terms of improving the response rate by a 50% reduction in monthly migraine days, Topiramate (RR = 50.06, 95% CI 3.18, 787.30) was the most effective; there was no statistically significant difference between all preventive drugs and placebo in ..." [2]

Drug	Indication	Dose / route (if sourced)	Key evidence
Pregabalin	reduction in migraine frequency	Not stated in the retrieved source for this use	"Compared with placebo, pregabalin (ratio of means [RoM], 0.38; 95% CI, 0.18-0.79) and topiramate with vitamin D3 (RoM, 0.44; 95% CI, 0.30-0.65) were associated with reduction in migraine frequency." [3]
Flunarizine	reduction in migraine frequency	Not stated in the retrieved source for this use	"Flunarizine (RoM, 0.46; 95% CI, 0.26-0.81), levetiracetam (RoM, 0.47; 95% CI, 0.30-0.72), riboflavin (RoM, 0.50; 95% CI, 0.32-0.77), cinnarizine (RoM, 0.64; 95% CI, 0.46-0.88), topiramate (RoM, 0.70; 95% CI, 0.55-0.89), and amitriptyline (RoM, 0.73; 95% CI, 0.54-0.97) were also associated with reduction in migraine frequency, but these findings were drawn from individual studies." [3]
Levetiracetam	reduction in migraine frequency	Not stated in the retrieved source for this use	"Flunarizine (RoM, 0.46; 95% CI, 0.26-0.81), levetiracetam (RoM, 0.47; 95% CI, 0.30-0.72), riboflavin (RoM, 0.50; 95% CI, 0.32-0.77), cinnarizine (RoM, 0.64; 95% CI, 0.46-0.88), topiramate (RoM, 0.70; 95% CI, 0.55-0.89), and amitriptyline (RoM, 0.73; 95% CI, 0.54-0.97) were also associated with reduction in migraine frequency, but these findings were drawn from individual studies." [3]
Riboflavin	reduction in migraine frequency	"Nov 13, 2025 · For migraine prophylaxis, the recommended dose of riboflavin is 400 mg daily as a single oral dose." source (from a retrieved page, not an FDA label)	"Flunarizine (RoM, 0.46; 95% CI, 0.26-0.81), levetiracetam (RoM, 0.47; 95% CI, 0.30-0.72), riboflavin (RoM, 0.50; 95% CI, 0.32-0.77), cinnarizine (RoM, 0.64; 95% CI, 0.46-0.88), topiramate (RoM, 0.70; 95% CI, 0.55-0.89), and amitriptyline (RoM, 0.73; 95% CI, 0.54-0.97) were also associated with reduction in migraine frequency, but these findings were drawn from individual studies." [3]
Cinnarizine	reduction in migraine frequency	Not stated in the retrieved source for this use	"Flunarizine (RoM, 0.46; 95% CI, 0.26-0.81), levetiracetam (RoM, 0.47; 95% CI, 0.30-0.72), riboflavin (RoM, 0.50; 95% CI, 0.32-0.77), cinnarizine (RoM, 0.64; 95% CI, 0.46-0.88), topiramate (RoM, 0.70; 95% CI, 0.55-0.89), and amitriptyline (RoM, 0.73; 95% CI, 0.54-0.97) were also associated with reduction in migraine frequency, but these findings were drawn from individual studies." [3]
amitriptyline	migraine prophylaxis (population: episodic migraine)	Not stated in the retrieved source for this use	"Results In episodic migraine with an average of 8 migraine or headache days/month at baseline, oral prophylactic drugs lowered the number of migraine days per month by 1.27 (beta-blockers), 0.44 (flunarizine), 1.2 (amitriptyline), and 1.4 (topiramate) compared to placebo." [4]

Drug	Indication	Dose / route (if sourced)	Key evidence
fremanezumab	migraine prophylaxis episodic/chronic	Not stated in the retrieved source for this use	"Monoclonal antibodies against calcitonin gene-related peptide (CGRP) or the CGRP receptor are effective against both episodic and chronic migraine: Eptinezumab lowered the number of migraine days per month by 0.7-3.2, fremanezumab by 1.3-3.8, galcanezumab by 1.1-3.7, and erenumab by 1.0-2.5." [4]
galcanezumab	migraine prophylaxis episodic/chronic	"Jun 24, 2026 · Galcanezumab is given as a monthly injection under the skin (subcutaneous injection) using a single-dose prefilled pen or a single-dose prefilled syringe after the first loading dose. 2 DOSAGE AND ADMINISTRATION 2.1 Recommended Dosing for Migraine The recommended dosage of EMGALITY ..." source (from a retrieved page, not an FDA label)	"Monoclonal antibodies against calcitonin gene-related peptide (CGRP) or the CGRP receptor are effective against both episodic and chronic migraine: Eptinezumab lowered the number of migraine days per month by 0.7-3.2, fremanezumab by 1.3-3.8, galcanezumab by 1.1-3.7, and erenumab by 1.0-2.5." [4]
erenumab	migraine prophylaxis episodic/chronic	Not stated in the retrieved source for this use	"Monoclonal antibodies against calcitonin gene-related peptide (CGRP) or the CGRP receptor are effective against both episodic and chronic migraine: Eptinezumab lowered the number of migraine days per month by 0.7-3.2, fremanezumab by 1.3-3.8, galcanezumab by 1.1-3.7, and erenumab by 1.0-2.5." [4]
atogepant	migraine prophylaxis episodic/chronic	"Apr 15, 2026 · For episodic migraine, start atogepant at 10 mg, 30 mg, or 60 mg once daily; for chronic migraine, use 60 mg once daily, with dose adjustments required for severe renal impairment (10 mg daily for episodic migraine only, avoid in chronic migraine) and avoidance in severe hepatic ..." source (from a retrieved page, not an FDA label)	"High-level evidence also supports the efficacy of the CGRP receptor antagonist atogepant (0.7-2.4 fewer migraine days per month) in both episodic and chronic migraine." [4]
Melatonin	as stated in the quoted source sentence	Not stated in the retrieved source for this use	"Versus placebo, melatonin reduced attack duration (MD -4.98 h; 95% CI -9.30 to -0.67; p = 0.02), headache days (MD -1.54 days; 95% CI -2.50 to -0.58; p Conclusions Melatonin demonstrates benefits over placebo for reducing migraine burden and improving patient-reported outcomes, with a favourable safety profile." [5]

Recently approved or updated (FDA, EMA and CDSCO announcements)

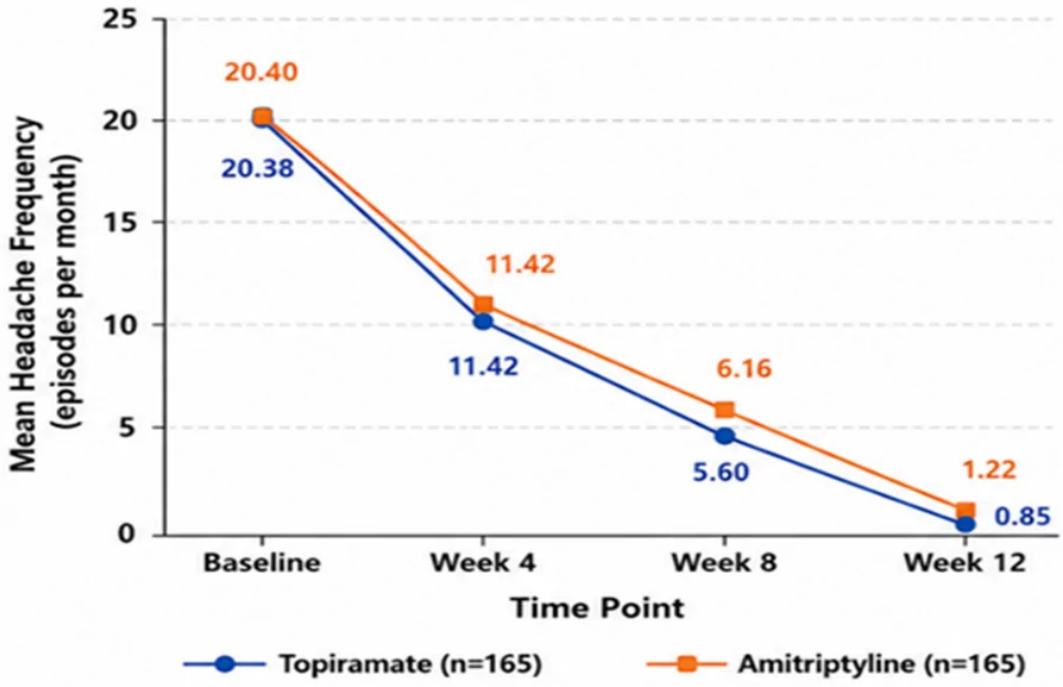
No FDA, EMA or CDSCO announcement dated within the last 3 years was found for this question. This does not mean none exists.

1 further regulator page(s) were read but are not listed: the page states no date, or a date more than 3 years old, so they cannot be shown as recent.

Safety warnings (FDA boxed warnings)

Drug(s) (FDA label)	Boxed warning, in the label's words
Riboflavin	"Accidental overdose of iron-containing products is a leading cause of fatal poisoning in children under 6. KEEP THIS PRODUCT OUT OF THE REACH OF CHILDREN. In case of accidental overdose, call a doctor or poison control center immediately."
amitriptyline	"Suicidality and Antidepressant Drugs: Antidepressants increased the risk compared to placebo of suicidal thinking and behavior (suicidality) in children, adolescents, and young adults in short-term studies of major depressive disorder (MDD) and other psychiatric disorders."
Valproic Acid	"Hepatotoxicity General Population: Hepatic failure resulting in fatalities has occurred in patients receiving valproate and its derivatives. These incidents usually have occurred during the first six months of treatment. Serious or fatal hepatotoxicity may be preceded by non-specific symptoms such as malaise, weakness, lethargy, facial edema, anorexia, and vomiting. In patients with epilepsy, a loss of seizure control may also occur."

These are the labels' own boxed warnings, not a complete safety review: read the full label. No boxed warning on the retrieved label of: Topiramate, Eptinezumab, Pregabalin, Levetiracetam, fremanezumab, galcanezumab, erenumab, atogepant, Melatonin. No FDA label could be matched for: Flunarizine, Cinnarizine.



Time Point	Topiramate (n=165) Mean ± SD	Amitriptyline (n=165) Mean ± SD
Baseline	20.38 ± 4.18	20.40 ± 4.28
Week 4	11.42 ± 2.84	11.42 ± 3.01
Week 8	5.60 ± 1.76	6.16 ± 2.06
Week 12	0.85 ± 0.63	1.22 ± 0.81

Lower values indicate fewer headache episodes.

SD: Standard Deviation

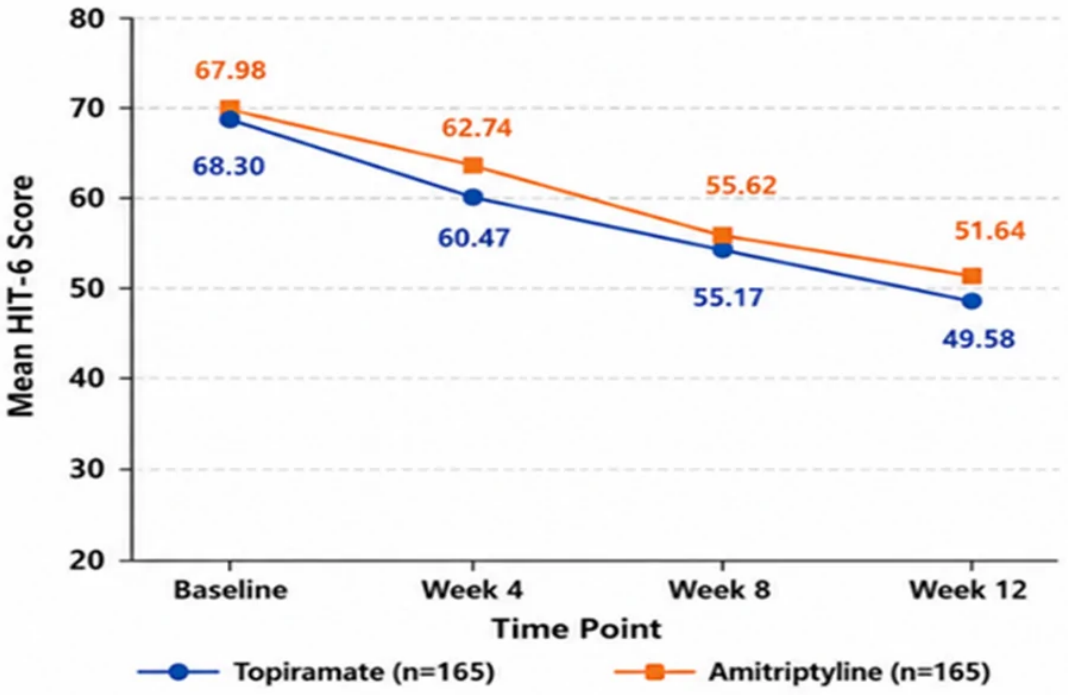
What this figure shows: Patients on Topiramate had fewer headache days per month (0.85) than those on Amitriptyline (1.22) by week 12, showing Topiramate's greater reduction in migraine frequency. This comparison helps evaluate which drug may offer better prophylaxis for chronic migraine patients.

How it helps with treatment options: This figure allows doctors to compare the relative effectiveness of Topiramate versus Amitriptyline in reducing migraine frequency over 12 weeks.

How to read this figure: The graph shows monthly headache frequency (y-axis) over time (x-axis) for two groups: Topiramate and Amitriptyline. Lower values on the y-axis mean fewer headache days per month.

Why this figure was chosen: The model's reason, from the caption: "Trend in headache frequency over time". Checks passed: the paper was judged directly relevant to the question; the article states an open licence; the model's five-part relevance test was passed (helps answer, clinically meaningful, adds to the report, not decorative, shows the kind of information the question asks for); the picture is large enough to read on the page; a vision model confirmed that the image matches its caption and is readable.

Figure 2. Trend in headache frequency over time.



Time Point	Topiramate (n=165) Mean ± SD	Amitriptyline (n=165) Mean ± SD
Baseline	68.30 ± 4.60	67.98 ± 4.29
Week 4	60.47 ± 5.38	62.74 ± 5.32
Week 8	55.17 ± 5.03	55.62 ± 5.46
Week 12	49.58 ± 6.16	51.64 ± 5.94

Lower scores indicate less headache-related disability.

HIT-6: Headache Impact Test-6; SD: Standard Deviation

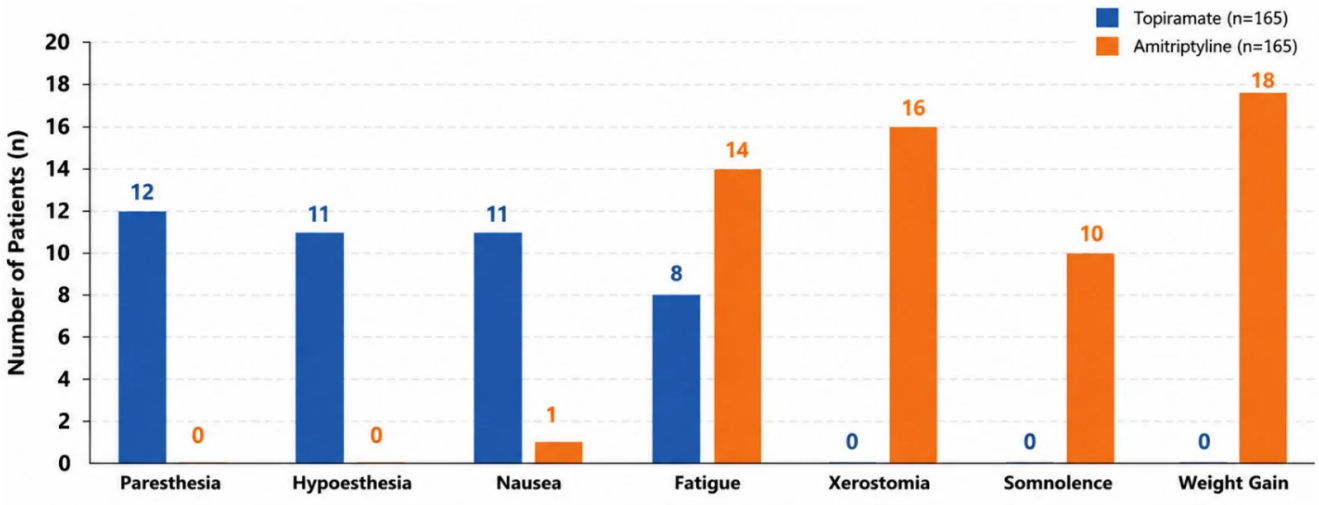
What this figure shows: This figure compares topiramate and amitriptyline for chronic migraine prophylaxis, showing both drugs reduced HIT-6 scores over 12 weeks, with topiramate yielding a lower score (49.58) than amitriptyline at week 12. It highlights how disability improved with both treatments, though topiramate appeared slightly more effective in this study.

How it helps with treatment options: This figure lets doctors compare the disability-reducing effects of topiramate versus amitriptyline over time, showing both treatments' relative performance in chronic migraine management.

How to read this figure: The graph shows how headache-related disability (measured by HIT-6 score) changes over 12 weeks for two groups: one treated with topiramate and the other with amitriptyline. A lower HIT-6 score means less disability due to headaches, and the lines track the average score decline over time.

Why this figure was chosen: The model's reason, from the caption: “Trend in HIT-6 score over time”. Checks passed: the paper was judged directly relevant to the question; the article states an open licence; the model's five-part relevance test was passed (helps answer, clinically meaningful, adds to the report, not decorative, shows the kind of information the question asks for); the picture is large enough to read on the page; a vision model confirmed that the image matches its caption and is readable.

Figure 3. Trend in HIT-6 score over time.



Adverse Event	Topiramate (n=165) n (%)	Amitriptyline (n=165) n (%)
Paresthesia	12 (7.3%)	0 (0.0%)
Hypoesthesia	11 (6.7%)	0 (0.0%)
Nausea	11 (6.7%)	1 (0.6%)
Fatigue	8 (4.8%)	14 (8.5%)
Xerostomia	0 (0.0%)	16 (9.7%)
Somnolence	0 (0.0%)	10 (6.1%)
Weight Gain	0 (0.0%)	18 (10.9%)

Values are presented as number and percentage of patients within each treatment group.

What this figure shows: This figure compares the safety profiles of topiramate and amitriptyline in chronic migraine patients, showing which adverse events were more common in each group. Topiramate had higher rates of paresthesia and fatigue, while amitriptyline showed more weight gain, relevant to weighing prophylaxis trade-offs.

How it helps with treatment options: This figure helps doctors compare the side effect profiles of topiramate and amitriptyline when choosing migraine prophylaxis, highlighting which adverse events are more likely with each drug.

How to read this figure: The figure compares the percentage of patients experiencing specific adverse events (e.g., paresthesia, fatigue, weight gain) for two migraine treatments: topiramate and amitriptyline. Higher bars indicate a greater proportion of patients reporting that adverse event.

Why this figure was chosen: The model's reason, from the caption: “Adverse events by treatment group”. Checks passed: the paper was judged directly relevant to the question; the article states an open licence; the model's five-part relevance test was passed (helps answer, clinically meaningful, adds to the report, not decorative, shows the kind of information the question asks for); the picture is large enough to read on the page; a vision model confirmed that the image matches its caption and is readable.

Figure 4. Adverse events by treatment group.

Clinical trials

Trial	Intervention	Phase	Status (last update)	What the record shows
NCT07871097	First dose rimegepant, second dose placebo; First dose placebo, second dose rimegepant	Phase 4	Recruiting (2026-10-13)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06810505	Atogepant; Placebo for Atogepant	Phase 3	Recruiting (2026-10-09)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07568496	Lidocaine 2% and Methylprednisolone 80 mg; Placebo	Phase not applicable	Recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07629635	Prasugrel 5mg; Transcatheter Closure of Patent Foramen Ovale	Phase 3	Recruiting (2026-10-08)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07849283	•Group A: Patients will receive Medical treatment only: Beta-blockers (Propranolol) , ...	Phase not applicable	Not yet recruiting (2026-10-07)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07816367	Elismetrep; Placebo	Phase 3	Recruiting (2026-10-07)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07861412	Patient Headache Diary; Patient Reported Outcome	Phase not applicable	Not yet recruiting (2026-10-06)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT05711394	Atogepant; Placebo-Matching Atogepant	Phase 3	Recruiting (2026-10-06)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT06792500	Glycerol Tributyrate	Phase 1/2	Not yet recruiting (2026-10-05)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT05983354	Oral metoclopramide and ibuprofen; Intravenous metoclopramide and ketorolac	Phase not applicable	Withdrawn (2026-10-05)	Registry status WITHDRAWN; no effectiveness conclusion.
NCT07616128	EVO756, Dose 1; EVO756, Dose 2	Phase 2	Recruiting (2026-10-02)	Under investigation; results not yet available; effectiveness cannot be inferred.
NCT07856056	Nortriptyline; Amitriptyline	Phase 4	Not yet recruiting (2026-10-02)	Under investigation; results not yet available; effectiveness cannot be inferred.

Registry status shows that research is being conducted, not that a treatment works. Registry status is separate from published results.

Key studies

1. **Pringsheim et al. (2026)**, Systematic Review: For episodic migraine, high-confidence evidence showed that galcanezumab and erenumab are more effective than placebo in reducing headache frequency. Moderate-confidence evidence supported benefit from atogepant, eptinezumab, fremanezumab, propranolol, topiramate, and valproate. For chronic migraine, high-confidence evidence supported reductions in headache frequency with fremanezumab, galcanezumab, and onabotulinumtoxinA [11].
2. **Diener et al. (2026)**, Review-Article: Oral prophylactic drugs lowered the number of migraine days per month by 1.27 (beta-blockers), 0.44 (flunarizine), 1.2 (amitriptyline), and 1.4 (topiramate) compared to placebo. Monoclonal antibodies against CGRP or the CGRP receptor reduced migraine days by 0.7–3.8, with atogepant showing 0.7–2.4 fewer migraine days per month [4].

3. **Talandashti et al. (2025)**, Meta-Analysis: Magnesium supplementation reduced migraine attacks by 2.51, severity by 0.88, and monthly migraine days by 1.66 compared to controls [8].

Conflicting evidence

No direct disagreement was retrieved between studies on migraine prophylaxis efficacy for established treatments (e.g., CGRP monoclonal antibodies or topiramate). However, **Tepper et al. (2025)** notes that melatonin has "very low certainty for evidence of efficacy," while **Abouelmagd et al. (2026)** reports statistically significant reductions in migraine burden with melatonin [12, 5]. The reason for this discrepancy is not established in the retrieved data.

What remains under investigation

- **Head-to-head comparisons**: Whether CGRP monoclonal antibodies (e.g., eptinezumab vs. fremanezumab) or gepants (e.g., atogepant vs. rimegepant) are superior in specific migraine subtypes (e.g., chronic vs. episodic) remains unanswered.
- **Long-term safety and adherence**: The retrieved data highlights gaps in long-term real-world safety and adherence for gepants and CGRP inhibitors, particularly in lower-resource settings.
- **Dietary supplements vs. pharmacotherapy**: While magnesium, riboflavin, and CoQ10 show promise in emerging evidence, their comparative efficacy and safety against established drugs (e.g., topiramate) are not yet established.
- **Sex-specific responses**: Observational evidence suggests sex differences in satisfaction with rimegepant [6], but efficacy differences by sex are not directly investigated in randomized trials.
- **Cost-effectiveness in diverse populations**: Economic evaluations [7] focus on high-income countries; cost-effectiveness in lower-resource settings or specific subgroups (e.g., older adults) is understudied.
- **Ultrasound-guided nerve blocks**: While proximal vs. distal greater occipital nerve blocks show efficacy [1], their role as a first-line or adjunctive therapy compared to pharmacotherapy is not yet defined.

Evidence limitations

- Findings are drawn from abstracts, registry records and web pages, not full texts; study quality was not appraised from the full papers.
- Doses appear only where an FDA label gives them for this condition (section 2.1); for any other use, see a prescribing source. Approval dates appear only where a retrieved source states them.
- 3 retrieved record(s) were excluded because the relevance reading judged them not to answer the question.
- The web search timed out, so it may be incomplete; the absence of evidence cannot be concluded from it.

General information from the retrieved literature; it is not a diagnosis or a personal treatment decision.

Sources

1. Sabet H, Samir A, Lashin B, et al. Exploring Proximal and Distal Ultrasound-Guided Greater Occipital Nerve Block Approaches in Migraine: A Systematic Review and Meta-Analysis. *Pain practice : the official journal of World Institute of Pain*. 2026-11-01. PMID 42743099. DOI 10.1111/papr.70202.
2. Zhao C, Li C, Yu X, et al. Effectiveness and safety of pharmacological prophylaxis for chronic migraine: a systematic review and network meta-analysis. *Journal of neurology*. 2024 Sep. PMID 38910144. DOI 10.1007/s00415-024-12512-z.
3. Kohandel Gargari O, Aghajanian S, Togha M, et al. Preventive Medications in Pediatric Migraine: A Network Meta-Analysis. *JAMA network open*. 2024 Oct 1. PMID 39388181. DOI 10.1001/jamanetworkopen.2024.38666.

4. Diener HC, Grans J, Reuter U, et al. Drugs for Migraine Prophylaxis. *Deutsches Arzteblatt international*. 2026-03-06. PMID 41572863. DOI 10.3238/arztebl.m2025.0234.
5. Abouelmagd ME, Aldemerdash MA, Khatatbeh AA, et al. Efficacy and Safety of Melatonin in Migraine Prophylaxis: A Systematic Review and Meta-Analysis of Randomized Controlled Trials. *Current pain and headache reports*. 2026-02-02. PMID 41627537. DOI 10.1007/s11916-025-01461-5.
6. Monteith T, Cirillo J, Nakajima K, et al. Sex differences in the real-world use of rimegepant for short-term prevention of migraine in the United States. *Frontiers in neurology*. 2026. PMID 42851312. DOI 10.3389/fneur.2026.1901123.
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Other sources (regulators, guideline organisations, web pages, FDA labels)

- dose page for Eptinezumab
- dose page for Riboflavin
- dose page for galcanezumab
- dose page for atogepant
- FDA drug label: Valproic Acid (label effective 2024-05-27)

Research support, not a substitute for clinical judgement. Some items could not be fully verified; the detail is in the saved verification record.

Figure credits

Figure 2: Comparative effectiveness and safety of topiramate versus amitriptyline for the prophylactic management of chronic migraine: a prospective comparative study. PMID 42851587, DOI 10.3389/fpain.2026.1923623, PubMed Central (PMC13646050). Not discussed in the report text. Licence: CC BY 4.0; original image, unaltered.

Figure 3: Comparative effectiveness and safety of topiramate versus amitriptyline for the prophylactic management of chronic migraine: a prospective comparative study. PMID 42851587, DOI 10.3389/fpain.2026.1923623, PubMed Central (PMC13646050). Not discussed in the report text. Licence: CC BY 4.0; original image, unaltered.

Figure 4: Comparative effectiveness and safety of topiramate versus amitriptyline for the prophylactic management of chronic migraine: a prospective comparative study. PMID 42851587, DOI 10.3389/fpain.2026.1923623, PubMed Central (PMC13646050). Not discussed in the report text. Licence: CC BY 4.0; original image, unaltered.

Each figure shows one study's result; it does not by itself establish that a treatment works or justify a recommendation.